

Marks-80

Time-3Hrs

A. Answer the following question with select the correct options.

1×45=45

1. The canal system is characteristic feature of

- (A) Arthropods
- (B) Mollusca
- (C) sponges
- (D) echinoderms

2. Cistron is

- (A) The coding sequence of DNA
- (B) The functional unit of DNA molecule that codes for a particular gene product
- (C) Intervening non coding sequence of DNA
- (D) The sequences which are removed during RNA splicing.

3. This is the largest phylum of Animal on the earth.

- (A) Mollusca
- (B) Amphibia
- (C) Arthropoda
- (D) Aves

4. The jawless vertebrate is

- (A) crocodile
- (B) zoris
- (C) *Hyla*
- (D) *Petromyzon*

5. Ribosomes are made up of

- (A) RNA only
- (B) RNA and Proteins
- (C) RNA, DNA and Proteins
- (D) nucleic acids, proteins and lipids

6. Which of the following snake is non-poisonous?

- (A) cobra
- (B) krait
- (C) viper
- (D) python

7. Which of the following is the function of lysosomes?

- (A) autophagy
- (B) autolysis
- (C) digestion
- (D) all of the above

8. Phylum of *Taenia Solium* is

- (A) Aschelminthes
- (B) Annelids
- (C) Platyhelminthes
- (D) mollusca

9. Where is DNA present in the eukaryotic cells?

- (A) Inside the nucleus
- (B) With other cellular contents
- (C) Inside the ribosomes
- (D) Not present

10. Which of the following nitrogenous base is not present in DNA?

- (A) Thymine
- (B) Adenine
- (C) Guanine
- (D) Uracil

11. The bases are held together in a DNA double helix by hydrogen bonds. These bonds are

- (A) Ionic bonds
- (B) Covalent bonds
- (C) Non-covalent bonds
- (D) Van der Waals forces

12. In a fetus, where are lymphocytes produced?

- (A) In the spleen
- (B) In the bone marrow
- (C) In the liver
- (D) In the heart

13. Anticodon is present in

- (A) DNA
- (B) tRNA
- (C) rRNA
- (D) mRNA

14. Name the secondary structure of tRNA?

- (A) Cloverleaf
- (B) L-shaped
- (C) Duplex
- (D) Triple Helix

15. Glycolysis is the conversion of

- (A) Fructose into phosphoenolpyruvate
- (B) Fructose into pyruvate
- (C) Glucose into phosphoenolpyruvate
- (D) Glucose into pyruvate

16. Which of the following rRNA molecules have peptidyl transferase activity in prokaryotes?

- (A) 23S rRNA
- (B) 28S rRNA

- (C) 5S Rrna
- (D) 18S rRNA

17. The percentage of human genome which encodes proteins is approximately

- (A) Less than 2%
- (B) 5%
- (C) 25%
- (D) 99%

18. In EMP pathway, the process by which ATP is formed from ADP is

- (A) reduction
- (B) oxidative phosphorylation
- (C) substrate-level phosphorylation
- (D) photo phosphorylation

19. Sickle cell anemia is caused

- (A) When valine is replaced by glutamic acid in beta polypeptide chain
- (B) When glutamic acid is replaced by valine in beta polypeptide chain
- (C) When glutamic acid is replaced by valine in alpha polypeptide chain
- (D) When valine is replaced by glutamic acid in alpha polypeptide chain

20. In prokaryotic cells, ribosomes are

- (A) 70 S
- (B) 80 S
- (C) 60S + 40S
- (D) 50S + 40S

21. Which of these is a characteristic of prokaryotic cells?

- (A) Absence of cell organelles
- (B) Absence of nucleus
- (C) Presence of 70S ribosomes
- (D) All of these

22. In prokaryotes, the hair-like outgrowths which attach to the surface of other bacterial cells are

- (A) Flagella
- (B) Pili
- (C) Capsule
- (D) Plasmids

23. B-cells and T-cells are two types of cells involved in _____

- (A) Innate Immunity
- (B) Active immunity
- (C) Passive immunity
- (D) Acquired immunity

24. Who coined the term mitochondria?

- (A) Kolliker
- (B) Benda
- (C) Fleenming
- (D) Robert Brown

25. On the Origin of Species was written by

- (A) Charles Darwin
- (B) Ludmila Kuprianova
- (C) Mikhail A. Fedonkin
- (D) None of the above

26. Which of the following cells of the immune system do not perform phagocytosis?

- (A) Macrophage
- (B) Neutrophil
- (C) Eosinophil
- (D) Basophil

27. The force that initiates evolution is _____

- (A) Variation
- (B) Mutation
- (C) Extinction
- (D) Adaptation

28. The limbless amphibian is

- (A) *Icthyophis*
- (B) *Hyla*
- (C) *Rana*
- (D) *Salamander*

29. The EMP pathway in eukaryotes usually takes place in

- (A) nucleus
- (B) lysosome
- (C) Golgi apparatus
- (D) cytosol

30. What is the first step in the payoff phase of glycolysis?

- (A) Reduction of 1, 3-bisphosphoglycerate to glyceraldehyde 3-phosphate
- (B) Oxidation of glyceraldehyde 3-phosphate to 1, 3-bisphosphoglycerate
- (C) Reversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate
- (D) Irreversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate

31. Plasmamembrane is

- (A) Permeable
- (B) Selectively permeable
- (C) Impermeable
- (D) Semi-permeable

32. In mitochondria cristae act as sites for
(A) proteinsynthesis
(B) phosphorylation of flavoproteins
(C) breakdown of macromolecules
(D) Oxidation-reductionreaction

33. Which of the following statements is true regarding the "law of segregation"?
(A) Law of segregation is the law of purity of genes
(B) Alleles separate from each other during gametogenesis
(C) Segregation of factors is due to the segregation of chromosomes during meiosis
(D) All of the above

34. How many phenotypes can occur in the human blood group ABO with alleles $I^A I^B i$?
(A) 2
(B) 3
(C) 4
(D) 1

35. Cystic fibrosis is
(A) Sex-linked recessive disorder
(B) Autosomal dominant disorder
(C) Autosomal recessive disorder
(D) Sex-linked dominant disorder

36. Alleles are
(A) Alternate forms of genes
(B) Linked genes
(C) Chromosomes that have crossed over
(D) Homologous chromosomes

37. The experiment that simulated conditions thought to be present on the early earth
(A) Hershey-Chase experiment
(B) Geiger-Marsden experiment
(C) Miller-Urey experiment
(D) Schiehallion experiment

38. Observation of species on _____ heavily inspired Darwin's theory of evolution.
(A) Ilha da Queimada Grande
(B) Guatemala
(C) Faroe Islands
(D) Galapagos Islands

39. Mammal's heart is
(A) Myogenic
(B) neurogenic
(C) voluntary
(D) sympathetic

40. What is the full form of PPLO?
(A) Pleural Parasite Like Organisms
(B) Phosphatic Pneumonia Like Organisms
(C) Pleuro Parasite Like Organisms
(D) Pleuro Pncumonia Like Organisms

41. Which of the following cells is involved in humoral immunity?
(A) T-cells
(B) B-cells
(C) Mast cells
(D) Both T and B cells

42. Body cavity lined by mesoderm is called
(A) coelenteron
(B) pseudocoel
(C) coelom
(D) blastocoel

43. Which of the following components of the vertebrate immune response occurs first upon invasion by a virus or bacterium?

(A) Activation of killer T lymphocytes
(B) Activation of B lymphocytes N
(C) The inflammatory response
(D) Mobilization of complement proteins

44. Tube feet are the locomotory organs of

(A) platyhelminthes
(B) Echinodermata
(C) Mollusca
(D) Arthropoda

45. Which of the following organelle is called "Suicidal Bag"?
(A) Mitochondria
(B) Endoplasmicreticulum
(C) Lysosome
(D) Ribosome

B. Answer any 10 questions from the following.

1. What is pseudocoelom? Where is it found?
2. What is Nucleotide?
3. In which cell organelle cristae is found? Mention its one function.
4. Mention two differences between Adaptive Immunity and Innate Immunity.
5. Write down the two characteristics features of Phylum Mollusca.
6. What do you mean by EMP pathway?
7. What is Central Dogma in molecular biology?
8. Mention one prokaryotic cell and one eukaryotic cell name.
9. What is tRNA? Mention its one function.
10. What is Nephridia? Where it is found?
11. What do you mean by allele?
12. What is Codon?

C. Answer any 3 questions from the following. 5×3=15

1. Describe the structure of DNA with suitable diagram.
2. What is Glycolysis? Write down the steps of Glycolysis.
3. Write down the short note on B-cell and T-cell.
4. Write down the characteristics features of Phylum Echinodermata. Mention two animal name under this phylum.